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1 Chats

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1.1 You

CM2526_exercises_09

PDF

Please solve the given exercise sheet. Give all exercise with detailed solution in LaTeX syntax.

1.2 Gemini

Show thinking

Here are the solutions to the exercises from the provided condensed matter physics sheet.

+1

Exercise 1: Wave equation in continuum

Problem Statement: Show that the N coupled equations of motion for a mono-atomic chain reduce to the continuum elastic wave equation for long wavelengths.

Solution: The equation of motion for the n -th atom is given by:

$$M \frac{d^2 u_n}{dt^2} = \sum_p C_p (u_{n+p} - u_n)$$

where u_n is the displacement of atom n and C_p are the force constants. For a mono-atomic chain, we assume symmetry such that $C_p = C_{-p}$.

1. Continuum Approximation For long wavelengths ($\lambda \gg a$), the displacement $u_n(t)$ changes slowly from atom to atom. We can treat the displacement as a continuous function $u(x, t)$, where $x = na$. We expand $u_{n+p} = u(x + pa)$ in a Taylor series around x :

$$u(x + pa) \approx u(x) + \frac{\partial u}{\partial x}(pa) + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} (pa)^2 + \dots$$

Substituting this into the term $(u_{n+p} - u_n)$:

$$u_{n+p} - u_n \approx pa \frac{\partial u}{\partial x} + \frac{1}{2} p^2 a^2 \frac{\partial^2 u}{\partial x^2}$$

2. Summation Substitute this back into the equation of motion:

$$M \frac{\partial^2 u}{\partial t^2} = \sum_p C_p \left(pa \frac{\partial u}{\partial x} + \frac{1}{2} p^2 a^2 \frac{\partial^2 u}{\partial x^2} \right)$$

Separating the terms:

$$M \frac{\partial^2 u}{\partial t^2} = \left(a \sum_p p C_p \right) \frac{\partial u}{\partial x} + \left(\frac{a^2}{2} \sum_p p^2 C_p \right) \frac{\partial^2 u}{\partial x^2}$$

3. Symmetry Argument Due to inversion symmetry of the Bravais lattice, the force constants satisfy $C_p = C_{-p}$. Therefore, the term linear in p vanishes:

$$\sum_p p C_p = 0$$

The equation reduces to:

$$M \frac{\partial^2 u}{\partial t^2} = \left(\sum_p \frac{1}{2} p^2 a^2 C_p \right) \frac{\partial^2 u}{\partial x^2}$$

4. Result Dividing by M , we obtain the continuum elastic wave equation:

$$\frac{\partial^2 u}{\partial t^2} = v_s^2 \frac{\partial^2 u}{\partial x^2}$$

where the speed of sound v_s is identified as:

$$v_s = \sqrt{\frac{1}{M} \sum_p \frac{1}{2} p^2 a^2 C_p}$$

Exercise 2: Linear chain with next nearest neighbour interactions

Problem Statement: A linear chain with nearest neighbor (NN) coupling C_1 and next-nearest neighbor (NNN) coupling C_2 . C_2 is ν times smaller than C_1 ($C_2 = C_1/\nu$).

- (a) Find the dispersion relation $\omega(k)$ The equation of motion including NN ($p = \pm 1$) and NNN ($p = \pm 2$) interactions is:

$$M \ddot{u}_n = C_1 (u_{n+1} + u_{n-1} - 2u_n) + C_2 (u_{n+2} + u_{n-2} - 2u_n)$$

Using the ansatz $u_n(t) = A e^{i(kna - \omega t)}$, we find:

$$-M\omega^2 = C_1 (e^{ika} + e^{-ika} - 2) + C_2 (e^{i2ka} + e^{-i2ka} - 2)$$

$$M\omega^2 = 2C_1 (1 - \cos(ka)) + 2C_2 (1 - \cos(2ka))$$

Using the identity $1 - \cos(x) = 2 \sin^2(x/2)$:

$$\omega^2(k) = \frac{4C_1}{M} \sin^2\left(\frac{ka}{2}\right) + \frac{4C_2}{M} \sin^2(ka)$$

Substituting $C_2 = C_1/\nu$:

$$\omega(k) = \sqrt{\frac{4C_1}{M} \left(\sin^2\left(\frac{ka}{2}\right) + \frac{1}{\nu} \sin^2(ka) \right)}$$

(b) For which integer values of ν does the maximum lie at $k < \pi/a$? To find the extrema, we take the derivative of ω^2 with respect to k and set it to zero:

$$\frac{d(\omega^2)}{dk} = \frac{2C_1 a}{M} \sin(ka) + \frac{2C_2(2a)}{M} \sin(2ka) = 0$$

Using $\sin(2ka) = 2 \sin(ka) \cos(ka)$:

$$\frac{2a}{M} \sin(ka) [C_1 + 4C_2 \cos(ka)] = 0$$

The solutions are $\sin(ka) = 0$ (at $k = 0, \pi/a$) or:

$$\cos(ka) = -\frac{C_1}{4C_2}$$

Substituting $C_2 = C_1/\nu$, the condition becomes:

$$\cos(ka) = -\frac{C_1}{4(C_1/\nu)} = -\frac{\nu}{4}$$

For a maximum to exist inside the Brillouin zone ($0 < k < \pi/a$), we must have $|\cos(ka)| < 1$. Thus:

$$\left| -\frac{\nu}{4} \right| < 1 \implies \nu < 4$$

Since ν is an integer (and typically $\nu \geq 1$ as C_2 is “smaller”), the values are: $\nu = 1, 2, 3$

(c) Maximum frequency for $\nu = 2$ For $\nu = 2$, the maximum occurs at $\cos(ka) = -2/4 = -1/2$. This corresponds to $ka = 2\pi/3$. Substitute this into the dispersion relation with $C_2 = C_1/2$:

$$\omega_{max}^2 = \frac{2C_1}{M} (1 - (-1/2)) + \frac{2(C_1/2)}{M} (1 - \cos(4\pi/3))$$

Note that $\cos(4\pi/3) = -1/2$.

$$\omega_{max}^2 = \frac{2C_1}{M} \left(\frac{3}{2} \right) + \frac{C_1}{M} (1 - (-0.5))$$

$$\omega_{max}^2 = \frac{3C_1}{M} + \frac{1.5C_1}{M} = \frac{4.5C_1}{M} = \frac{9C_1}{2M}$$

$$\omega_{max} = 3\sqrt{\frac{C_1}{2M}}$$

This is higher than the frequency at the boundary $\omega(\pi/a) = \sqrt{4C_1/M} = 2\sqrt{C_1/M}$.

(d) Speed of sound v_s The speed of sound is the group velocity at $k \rightarrow 0$. For small k , $\sin(x) \approx x$.

$$\omega^2 \approx \frac{4C_1}{M} \left(\frac{ka}{2} \right)^2 + \frac{4C_2}{M} (ka)^2$$

$$\omega^2 \approx \frac{C_1 a^2 k^2}{M} + \frac{4C_2 a^2 k^2}{M} = \frac{a^2 k^2}{M} (C_1 + 4C_2)$$

Taking the square root:

$$\omega \approx \sqrt{\frac{C_1 + 4C_2}{M}} ak$$

Thus,

$$v_s = \left. \frac{d\omega}{dk} \right|_{k \rightarrow 0} = a \sqrt{\frac{C_1 + 4C_2}{M}}$$

Exercise 3: Atomic chain of two types of atoms

Problem Statement: Diatomic chain, masses M_1 and M_2 . M_1 at $x_n = na$, M_2 at $x_n + d$ ($d < a$). Nearest neighbor coupling D .

(a) Differential equations Let $u_{1,n}$ be the displacement of mass M_1 in cell n , and $u_{2,n}$ for mass M_2 . The NN interactions for M_1 are with M_2 in the same cell (dist d) and M_2 in the previous cell (dist $a - d$). Since the coupling constant is D for both:

$$M_1 \ddot{u}_{1,n} = D(u_{2,n} - u_{1,n}) + D(u_{2,n-1} - u_{1,n})$$

$$M_2 \ddot{u}_{2,n} = D(u_{1,n} - u_{2,n}) + D(u_{1,n+1} - u_{2,n})$$

(b) Dispersion relation and BZ Sketch Using the ansatz $\sqrt{M_\alpha} u_{\alpha,n} = A_\alpha e^{i(kna - \omega t)}$: Substitute into EOMs:

$$M_1(-\omega^2) \frac{A_1}{\sqrt{M_1}} = D \left(\frac{A_2}{\sqrt{M_2}} - \frac{A_1}{\sqrt{M_1}} \right) + D \left(\frac{A_2}{\sqrt{M_2}} e^{-ika} - \frac{A_1}{\sqrt{M_1}} \right)$$

$$M_2(-\omega^2) \frac{A_2}{\sqrt{M_2}} = D \left(\frac{A_1}{\sqrt{M_1}} - \frac{A_2}{\sqrt{M_2}} \right) + D \left(\frac{A_1}{\sqrt{M_1}} e^{ika} - \frac{A_2}{\sqrt{M_2}} \right)$$

This simplifies to the linear system:

$$\left(\frac{2D}{M_1} - \omega^2 \right) A_1 - \frac{D}{\sqrt{M_1 M_2}} (1 + e^{-ika}) A_2 = 0$$

$$-\frac{D}{\sqrt{M_1 M_2}} (1 + e^{ika}) A_1 + \left(\frac{2D}{M_2} - \omega^2 \right) A_2 = 0$$

Setting the determinant to zero:

$$\left(\frac{2D}{M_1} - \omega^2 \right) \left(\frac{2D}{M_2} - \omega^2 \right) - \frac{D^2}{M_1 M_2} |1 + e^{-ika}|^2 = 0$$

Using $|1 + e^{-ika}|^2 = 2 + 2 \cos(ka) = 4 \cos^2(ka/2) = 4 - 4 \sin^2(ka/2)$:

$$\omega^4 - 2D \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \omega^2 + \frac{4D^2}{M_1 M_2} \sin^2 \left(\frac{ka}{2} \right) = 0$$

Solving for ω^2 :

$$\omega^2 = D \left(\frac{1}{M_1} + \frac{1}{M_2} \right) \pm D \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2} \sin^2 \left(\frac{ka}{2} \right)}$$

+ sign: Optical branch.

– sign: Acoustic branch.

Sketch: The first Brillouin zone is $[-\pi/a, \pi/a]$. The acoustic branch starts at 0, optical starts at a finite max value. They approach a gap at the zone boundary $\pm\pi/a$.

(c) Group velocity $v_g = d\omega/dk$.

Acoustic: v_g is constant (sound speed) near $k = 0$, decreases to 0 at $k = \pm\pi/a$.

Optical: v_g is 0 at $k = 0$, increases, then decreases back to 0 at $k = \pm\pi/a$.

(d) Solutions for limits

1. $ka \ll 1$ (Center of zone)

Acoustic: $\omega \approx v_s k$. $A_1 \approx \sqrt{M_1/M_2} A_2$. More physically $u_1 \approx u_2$. The atoms in the cell move in phase (sound wave).

Optical: $\omega \approx \sqrt{2D(1/M_1 + 1/M_2)}$. $M_1 u_1 + M_2 u_2 \approx 0$. The atoms move out of phase against each other (center of mass stationary).

2. $ka = \pm\pi$ (Zone boundary) $\sin^2(ka/2) = 1$. The term in the square root becomes $(1/M_1 - 1/M_2)^2$.

Optical: $\omega^2 = 2D(1/M_1 + 1/M_2) + 2D|1/M_1 - 1/M_2|$. Assume $M_1 < M_2$. Then $\omega^2 = 2D(1/M_1 + 1/M_2 + 1/M_1 - 1/M_2) = \frac{4D}{M_1}$.

Frequency depends on the lighter mass.

Motion: Heavier mass is stationary ($A_{heavy} = 0$), Lighter mass oscillates.

Acoustic: $\omega^2 = 2D/M_2$ (if $M_1 < M_2$).

Frequency depends on the heavier mass.

Motion: Lighter mass is stationary ($A_{light} = 0$), Heavier mass oscillates.

(e) Case $M_1 \gg M_2$ If M_1 is very heavy:

Optical Branch: $\omega \approx \sqrt{2D/M_2}$. The light atoms (M_2) oscillate rapidly while heavy atoms (M_1) are effectively stationary walls. The band is flat (Einstein oscillator model).

Acoustic Branch: ω is small, governed by the total mass or the heavy mass near the boundary.

(f) Numerical Calculation for NaCl

$M_{Na} \approx 23$ u (let M_1), $M_{Cl} \approx 35.5$ u (let M_2).

1 u $\approx 1.66 \times 10^{-27}$ kg.

$D = 20$ N/m.

1. Width of Optical Branch ($\Delta\omega$)

$$\omega_{opt}(0) = \sqrt{2D \left(\frac{1}{M_1} + \frac{1}{M_2} \right)}$$

$$\omega_{opt}(\pi/a) = \sqrt{\frac{2D}{M_1}} \quad (\text{since } M_1 < M_2)$$

Converting masses to kg: $M_1 \approx 3.82 \times 10^{-26}$ kg, $M_2 \approx 5.89 \times 10^{-26}$ kg.

$$\omega_{opt}(0) = \sqrt{40 \left(\frac{1}{3.82} + \frac{1}{5.89} \right) \times 10^{26}} \approx 4.15 \times 10^{13} \text{ rad/s}$$

$$\omega_{opt}(\pi/a) = \sqrt{\frac{40}{3.82 \times 10^{-26}}} \approx 3.24 \times 10^{13} \text{ rad/s}$$

$$\Delta\omega_{opt} = (4.15 - 3.24) \times 10^{13} = 0.91 \times 10^{13} \text{ rad/s}$$

2. Energy Gap Gap is between min of optical and max of acoustic. Max acoustic is at boundary: $\omega_{ac}(\pi/a) = \sqrt{2D/M_2}$.

$$\omega_{ac}(\pi/a) = \sqrt{\frac{40}{5.89 \times 10^{-26}}} \approx 2.61 \times 10^{13} \text{ rad/s}$$

Frequency Gap $\Delta\omega_{gap} = 3.24 \times 10^{13} - 2.61 \times 10^{13} = 0.63 \times 10^{13} \text{ rad/s}$. Energy Gap $E_g = \hbar\Delta\omega_{gap}$.

$$E_g \approx (6.582 \times 10^{-16} \text{ eV} \cdot \text{s}) \times (0.63 \times 10^{13} \text{ s}^{-1}) \approx 4.1 \text{ meV}$$

$$\text{Frequency Gap} \approx 6.3 \times 10^{12} \text{ rad/s}$$

Next Step: Would you like me to visualize the dispersion relation for Exercise 2 to verify the condition $\nu < 4$?

Sources
